SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10% **Total Marks:** 10

CASE STUDY: QUESTIONS AND ANSWERS

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The following observations are made:

• The team needs to embed a promotional video on the homepage.

• A registration form must securely send user data to the server.

• The navigation menu must follow semantic HTML standards.

• XML configuration files are being used but contain syntax errors.

• The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

QUESTIONS AND ANSWERS

Question: 1. Select the correct HTML tag for video embedding and justify your answer.

Answer: Correct option: B. <video src="video.mp4" controls></video>. The HTML5 <video> element is the standard element for embedding video content in a web page. The src attribute specifies the video file, while controls displays the browser's playback controls, such as play, pause, and volume. The other options use non-standard tags: <media>, <movie>, and <vid> are not standard HTML elements for video embedding.

Question: 2. Identify the appropriate HTTP method for form submission and explain why.

Answer: Correct option: B. POST. POST sends form data in the request body and is generally suitable for registration forms that create or submit data to the server. GET places submitted values in the URL query string and should not be used for sensitive information. POST alone does not encrypt data; HTTPS is required to protect information in transit.

Question: 3. Choose the proper semantic element for navigation and justify its usage.

Answer: Correct option: B. <nav>. The <nav> element identifies a section of a page that contains major navigation links. It gives the document clearer semantic structure and helps assistive technologies identify navigation areas. A is generic, <section> is for a thematic grouping of content, and <header> is introductory content or a container for headings and related material.

Question: 4. Identify the correct XML syntax and explain the error in other options.

Answer: Correct option: B. <note><to>User</to><from>Admin</from></note>. This is well-formed XML because each element is properly opened and closed, and the elements are nested correctly. Option A is missing the closing </note> tag. Option C mismatches the tags: <to> is closed with </from>. Option D is missing the closing angle bracket (>) in the closing tag, and the <note> element is not properly closed. XML requires properly nested, matching tags and a single root element.

Question: 5. Choose the correct protocol for web communication and justify your choice.

Answer: Correct option: B. HTTP. Hypertext Transfer Protocol (HTTP) is used for communication between web clients, such as browsers, and web servers to request and deliver web resources. HTTPS is the secure version of HTTP and should be used for modern websites because it protects data in transit using encryption. FTP is used for file transfer, SMTP for email transmission, and SNMP for network management.

CODE OF CONDUCT

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____